

Supplementary Material for CauSample

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A Proof of the Causal Sampling Bound

We restate Theorem 4.2 and establish its causal sampling bound through the three steps outlined below.

THEOREM A.1 (CAUSAL SAMPLING BOUND). *For a batch $X_i \in \mathbb{R}^{M \times T}$ and its SCS $\mathcal{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_K\}$, assume each child is an exact linear combination of its parents, whose vectors are linearly independent and span a subspace with incoherence at most μ . Let $\tau = \max_m |Pa(m)| \geq 1$, $l = \max_k |Root_k|$, and $\delta \in (0, 1)$. Then:*

- (1) *Given exact parent data, any child metric can be reconstructed exactly and its causal coefficients recovered with probability at least $1 - \delta$ by uniformly sampling d time steps without replacement, where $d \geq c\mu\tau \log(Kl/\delta)$ for a universal constant c ; if this requirement exceeds T , all entries are sampled.*
- (2) *If $Kl \leq r = \text{rank}(X_i)$, the causal bound is no larger than the global-rank bound under the same incoherence bound: $c\mu\tau \log(Kl/\delta) \leq c\mu r \log(r/\delta)$.*

Proof sketch. The proof establishes how the SCS supports matrix completion through three key steps:

- (1) **Characterize the local rank** (Lemma A.2): show that a child and its parents form a matrix whose SVD rank equals the causal rank, and relate this rank to the DAG's roots and the global rank.
- (2) **Prove that sampling preserves this rank** (Proposition A.3): apply a matrix Chernoff bound to obtain a sufficient sample count for retaining the local rank.
- (3) **Establish recovery in the causal graph** (Lemma A.4): show that rank-preserving samples uniquely determine the causal coefficients and recover the child, and distinguish this case from vectors outside the parent span.

We then combine these results to prove the theorem's sampling bound and its comparison with the global-rank bound. After the theorem proof, Corollary A.6 extends the recovery argument to all children by a union bound and induction along the causal orders.

Notation. For a child metric m in batch i , write $P_m = X_i(Pa(m), :) \in \mathbb{R}^{\tau_m \times T}$, $\tau_m = |Pa(m)|$, and $C_m = [P_m^\top, x_{m,i}^\top]$. The theorem assumes $x_{m,i} = w_m P_m$ with linearly independent parent rows; coefficients are fixed within this batch but may change between batches. Let $l_k = |Root_k|$, $l = \max_k l_k$, and $r_k = \text{rank}(X_i(\pi_k, :))$. For $q \geq 1$, an orthonormal basis $U \in \mathbb{R}^{T \times q}$ has incoherence $\mu(U) = (T/q) \max_t \|U^\top e_t\|_2^2$, where e_t is a standard basis vector. The bound μ covers all parent row spaces and, for the comparison, the global row space. Sampling selects distinct time steps uniformly without replacement, independently of the fixed data. Following the main paper, $\Omega_m \in \{0, 1\}^{1 \times T}$ records the sampled time steps; with $D_m = \text{diag}(\Omega_m)$, $x_{m,i,\Omega_m} = x_{m,i} D_m$ and the sampled parent matrix is $P_m D_m$. All logarithms are natural.

Step 1: relate causal rank to the local SVD rank. We first show that the causal rank $\tau_m = |Pa(m)|$ equals the SVD rank of the submatrix containing metric m and its parents.

LEMMA A.2 (CAUSAL RANK EQUALS THE LOCAL SVD RANK). *For a child metric m in $\mathcal{G}_k$, let $B_m = [P_m; \mathbf{x}_{m,i}] = C_m^\top \in \mathbb{R}^{(\tau_m+1) \times T}$ be the corresponding row submatrix of X_i . Under the theorem's assumptions, B_m and C_m each have exactly τ_m positive singular values; their SVD ranks therefore equal the causal rank τ_m . Moreover,*

$$\tau_m \leq r_k \leq l_k \leq l, \quad \tau \leq \max_k r_k \leq r. \quad (1)$$

PROOF. The parent rows are linearly independent, so $\text{rank}(P_m) = \tau_m$. Since P_m is a row submatrix of B_m , $\text{rank}(B_m) \geq \tau_m$. Conversely, $\mathbf{x}_{m,i} = \mathbf{w}_m P_m$ makes every row of B_m a linear combination of the parent rows, giving $\text{rank}(B_m) \leq \tau_m$. Thus $\text{rank}(B_m) = \tau_m$. Matrix rank equals the number of positive singular values, and transposition preserves these values; hence

$$\#\{j : \sigma_j(B_m) > 0\} = \text{rank}(B_m) = \text{rank}(C_m) = \tau_m. \quad (2)$$

This identifies the structural parent count with the local SVD rank, so the sampling analysis can use τ_m without computing a global SVD.

The matrix B_m is a row submatrix of $X_i(\pi_k, :)$, which is itself a row submatrix of X_i , giving $\tau_m \leq r_k \leq r$. Every row in a DAG also lies in the span of its roots: starting from the roots, substitute each child's parent representation in causal order. Consequently, $r_k \leq l_k \leq l$, and taking maxima proves Equation (1). The local rank equality uses the theorem's linear independence assumption on the parent data. $\square$

Step 2: preserve the local rank under sampling. Sampling a child and its parents at the same time steps selects columns of B_m . We show that sufficiently many random columns preserve its SVD rank, following the rank-preservation analysis of subspace-based matrix completion [1]. The proposition is stated for rank q ; the local causal matrix has $q = \tau_m$ by Lemma A.2.

PROPOSITION A.3 (RANK PRESERVATION BY RANDOM SAMPLING). *Let $B \in \mathbb{R}^{p \times T}$ have SVD rank $q \geq 1$ and row-space incoherence at most μ . Let Ω select d distinct columns uniformly without replacement and set $D = \text{diag}(\Omega)$. For $0 < \varepsilon < 1$, if*

$$d \geq \min\{T, \lceil 8\mu q \log(q/\varepsilon) \rceil\}, \quad (3)$$

then $\text{rank}(BD) = q$ with probability at least $1 - \varepsilon$. Here BD keeps the sampled columns and zeros the others, so its rank equals that of the selected-column matrix.

PROOF. Full sampling preserves rank deterministically; assume $d < T$.

(i) *Construct the sampled Gram matrix.* Take a compact SVD $B = L\Sigma V^\top$ and let $Z = V^\top \in \mathbb{R}^{q \times T}$, whose rows are orthonormal. Since $L\Sigma$ has full column rank, $\text{rank}(BD) = \text{rank}(ZD)$. Writing $z_t = Z(:, t)$, define the sampled matrix $S = ZD = \sum_{t=1}^T \Omega(t) z_t e_t^\top$ and its Gram matrix

$$Y = SS^\top = \sum_{t=1}^T \Omega(t) z_t z_t^\top. \quad (4)$$

Because $\lambda_{\min}(Y) = \sigma_q(S)^2$, showing Y is positive definite proves that the sampled matrix retains rank q .

(ii) *Bound the expectation and individual summands.* Each column is included with probability d/T , and $ZZ^\top = I_q$, so

$$\mathbb{E}[Y] = \frac{d}{T} ZZ^\top = \frac{d}{T} I_q, \quad a = \lambda_{\min}(\mathbb{E}[Y]) = \frac{d}{T}. \quad (5)$$

For each column, incoherence gives

$$\lambda_{\max}(\mathbf{z}_t \mathbf{z}_t^\top) = \|\mathbf{z}_t\|_2^2 \leq L := \max_s \|\mathbf{z}_s\|_2^2 \leq \frac{\mu q}{T}. \quad (6)$$

These are the two quantities required by the matrix Chernoff bound.

(iii) Apply the matrix Chernoff bound. Lemma A.5, with $\eta = 1/2$, yields

$$\Pr\left\{\lambda_{\min}(\mathbf{Y}) \leq \frac{d}{2T}\right\} \leq q \exp\left(-\frac{d}{8TL}\right) \leq q \exp\left(-\frac{d}{8\mu q}\right) \leq \varepsilon. \quad (7)$$

The last inequality follows from Equation (3). With probability at least $1 - \varepsilon$, $\lambda_{\min}(\mathbf{Y}) > d/(2T) > 0$, so $\sigma_q(\mathbf{S}) > 0$ and $\text{rank}(\mathbf{B}\mathbf{D}) = \text{rank}(\mathbf{S}) = q$.

Applying this result to $\mathbf{B}_m$ gives the local requirement $d \geq c\mu\tau_m \log(\tau_m/\varepsilon)$, before rounding and full-sampling capping, for a sufficiently large universal constant c . The numerical factor in Equation (3) is one sufficient choice from this concentration argument; the theorem retains c rather than fixing its value. $\square$

Step 3: establish recovery in the causal graph. Rank preservation allows the samples to determine the causal coefficients uniquely. The following lemma proves this recovery property and examines the three cases according to whether a vector lies in the parent span, the DAG's root span, or neither. Only the parent-span case is needed for the theorem; the other two explain residual detection when the assumed representation does not hold.

LEMMA A.4 (CAUSAL RECOVERY AND RESIDUAL DETECTION). *Fix a DAG $\mathcal{G}_k$ and an exact parent matrix $\mathbf{P}_m$ of rank $\tau_m \geq 1$. Let $\mathcal{R}_k$ be the span of the DAG's root rows, so the row space of $\mathbf{P}_m$ is contained in $\mathcal{R}_k$. For a candidate row $\mathbf{x} \in \mathbb{R}^{1 \times T}$, define $\mathbf{B} = [\mathbf{P}_m; \mathbf{x}]$, $q = \text{rank}(\mathbf{B})$, and let μ_B denote its row-space incoherence. For $0 < \varepsilon < 1$, suppose $d \geq \min\{T, \lceil 8\mu_B q \log(q/\varepsilon) \rceil\}$ time steps are sampled uniformly without replacement. With probability at least $1 - \varepsilon$, the following conclusion holds for the applicable case:*

- (1) *If $\mathbf{x} \in \mathcal{R}_k$ but $\mathbf{x}$ is outside the row space of $\mathbf{P}_m$, then $\text{rank}(\mathbf{B}\mathbf{D}) = \tau_m + 1$ and the sampled least-squares residual is strictly positive.*
- (2) *If $\mathbf{x} = \mathbf{w}\mathbf{P}_m$, then $\text{rank}(\mathbf{B}\mathbf{D}) = \tau_m$; the sampled data uniquely determine $\mathbf{w}$ and reconstruct $\mathbf{x}$.*
- (3) *If $\mathbf{x} \notin \mathcal{R}_k$, then $\text{rank}(\mathbf{B}\mathbf{D}) = \tau_m + 1$ and the sampled least-squares residual is strictly positive.*

PROOF. Write $\mathbf{A} = \mathbf{P}_m \mathbf{D}$, $\mathbf{y} = \mathbf{x} \mathbf{D}$, and $\widehat{\mathbf{w}} = \mathbf{y} \mathbf{A}^\dagger$. Proposition A.3 gives the event $\text{rank}(\mathbf{B}\mathbf{D}) = \text{rank}(\mathbf{B}) = q$ with probability at least $1 - \varepsilon$.

Case 1: inside the DAG span but outside the parent span. Appending $\mathbf{x}$ increases rank from τ_m to $\tau_m + 1$. On the rank-preservation event, $\mathbf{y}$ cannot lie in the row space of $\mathbf{A}$, whose rank is at most τ_m . Consequently, $\|\mathbf{y} - \widehat{\mathbf{w}}\mathbf{A}\|_2^2 > 0$. Here the sampled matrix must preserve the augmented rank $q = \tau_m + 1$; its incoherence need not equal that of the parents alone.

Case 2: inside the parent span. The identity $\mathbf{x} = \mathbf{w}\mathbf{P}_m$ implies $\mathbf{y} = \mathbf{w}\mathbf{A}$ for every sampling vector, so the sampled residual is always zero. Since $\mathbf{B}\mathbf{D} = [\mathbf{A}; \mathbf{w}\mathbf{A}]$, rank preservation also makes $\mathbf{A}$ full row rank. Its pseudoinverse is therefore

$$\mathbf{A}^\dagger = \mathbf{A}^\top (\mathbf{A}\mathbf{A}^\top)^{-1}, \quad \mathbf{A}\mathbf{A}^\dagger = \mathbf{I}_{\tau_m}. \quad (8)$$

It follows that $\widehat{\mathbf{w}} = \mathbf{w}\mathbf{A}\mathbf{A}^\dagger = \mathbf{w}$, and hence

$$\widehat{\mathbf{x}} = \mathbf{x} \mathbf{D} (\mathbf{P}_m \mathbf{D})^\dagger \mathbf{P}_m = \mathbf{w} \mathbf{P}_m = \mathbf{x}. \quad (9)$$

This is the coefficient recovery used in Equation (7) of the main paper. In this case $q = \tau_m$ and the augmented and parent row spaces coincide, so $\mu_B \leq \mu$.

Case 3: outside the DAG span. Because the parent row space is contained in $\mathcal{R}_k$, $\mathbf{x} \notin \mathcal{R}_k$ also implies that $\mathbf{x}$ is outside the parent row space. The augmented matrix therefore has rank $\tau_m + 1$,

and the same argument as in Case 1 gives a positive sampled residual. For an arbitrary fixed outlier, this conclusion requires sampling to preserve the augmented rank; it is not a probability-one claim.

The residual statements use a zero threshold in exact arithmetic. A positive practical tolerance additionally requires a lower bound on the residual magnitude. Cases 1 and 3 detect failure of the specified representation; neither alone establishes that the metric is a root of the SCS. $\square$

PROOF OF THEOREM A.1. (1) Sampling bound and reconstruction. We combine the local-rank, rank-preservation, and recovery results above, first within one DAG and then across the K DAGs. Throughout, c denotes a sufficiently large universal constant that absorbs the numerical factor in Proposition A.3.

Consider a child m in $\mathcal{G}_k$. Lemma A.2 identifies its local SVD rank as τ_m . Proposition A.3 preserves this rank with probability at least $1 - \varepsilon$ using $d \geq c\mu\tau_m \log(\tau_m/\varepsilon)$ samples. On this event, Case 2 of Lemma A.4 uniquely recovers the causal coefficients and reconstructs the child from its exact parent data. Since $\tau_m \leq l_k$ and $\tau_m \leq \tau$, a common sufficient budget for any fixed child in this DAG is

$$c\mu\tau_m \log(\tau_m/\varepsilon) \leq c\mu\tau \log(l_k/\varepsilon). \quad (10)$$

For the K DAGs, assign failure tolerance $\varepsilon = \delta/K$ to each such recovery event. Using $l_k \leq l$ gives the claimed requirement:

$$c\mu\tau_m \log(K\tau_m/\delta) \leq c\mu\tau \log(Kl_k/\delta) \leq c\mu\tau \log(Kl/\delta). \quad (11)$$

Each fixed child therefore succeeds with probability at least $1 - \delta/K \geq 1 - \delta$, as required by the theorem. A union bound also gives joint probability at least $1 - \delta$ for at most one predetermined child per DAG. The guarantee for all children, including recovery of their parents, is proved in Corollary A.6. Rounding the budget upward preserves the result; if it exceeds T , full sampling gives deterministic recovery. Keeping τ_m instead of its maximum τ also yields the per-child bound used in Equation (8) of the main paper.

(2) Comparison with the global-rank bound. Balcan and Zhang [1, Theorem 4] derive a per-column sampling parameter d of order $\mu r \log(r/\delta)$ based on the global rank, under their stated model and sampling assumptions. We compare the sufficient budgets using the same incoherence upper bound and a common sufficiently large constant c . Our advantage is that the rank factor depends on local causal representations rather than all variations in the batch. Specifically, $C_m^\top$ is a row submatrix of $X_i(\pi_k, :)$, which is itself a row submatrix of X_i . Lemma A.2 therefore gives $\tau_m \leq r_k \leq r$ and $\tau \leq \max_k r_k \leq r$. Under the theorem's condition $Kl \leq r$,

$$c\mu\tau \log(Kl/\delta) \leq c\mu\tau \log(r/\delta) \leq c\mu r \log(r/\delta). \quad (12)$$

When $\tau < r$, the main reduction comes from this smaller causal rank: each child uses only its parents, whereas the global rank also counts independent variations elsewhere in the batch. Strict rank reduction holds when every parent span is a proper subspace of the global row space. For example, if $K \geq 2$ and the DAG row spaces are nonzero and mutually linearly independent, then $\tau \leq \max_k r_k < \sum_k r_k = r$. When $\tau < r$ and $Kl \leq r$, Equation (12) gives a strictly smaller sufficient budget before rounding and capping at T ; in general, only $\tau \leq r$ is guaranteed.

For $K = 1$, the bound reduces to $c\mu\tau \log(l/\delta)$, and the comparison holds under $l \leq r$. For $K = M$, all DAGs are singletons, so there are no child metrics; the framework uses temporal SMC for every metric. This proves both statements of Theorem A.1. $\square$

LEMMA A.5 (MATRIX CHERNOFF LOWER-TAIL BOUND). Let Y be a sum of positive semidefinite $q \times q$ matrices, either independent or drawn uniformly without replacement from a fixed finite collection. Suppose each summand has maximum eigenvalue at most $L > 0$, and write $a = \lambda_{\min}(\mathbb{E}[Y]) > 0$. For

$0 < \eta < 1$,

$$\Pr\{\lambda_{\min}(Y) \leq (1 - \eta)a\} \leq q \left(\frac{e^{-\eta}}{(1 - \eta)^{1-\eta}} \right)^{a/L} \leq qe^{-a\eta^2/(2L)}. \quad (13)$$

PROOF. For independent summands this is the matrix Chernoff inequality of Tropp [3, Corollary 5.2 and Remark 5.3]. For uniform sampling without replacement, let $\tilde{Y}$ be the sum of the same number of independent uniform draws with replacement from the same collection. The two sums have the same expectation and summand bound L . Gross and Nesme [2] show that $\mathbb{E} \text{tr} \exp(-\theta Y) \leq \mathbb{E} \text{tr} \exp(-\theta \tilde{Y})$ for every $\theta > 0$. Applying the same exponential-moment argument therefore gives Equation (13) without replacement as well. $\square$

Whole-batch recovery. The theorem above concerns a child given exact parent data. To recover a complete batch, Algorithm 1 processes each DAG in causal order, fully collects roots, and then fits each child from sampled entries and reconstructed parents. The root cost is separate from the child budget. The algorithm returns FAILURE if a sampled parent matrix loses rank; the following result bounds this probability jointly over all children.

Algorithm 1 Offline Causal Matrix Completion

Require: Query access to entries of $X_i \in \mathbb{R}^{M \times T}$; SCS $\mathcal{G}$ with orders π_k and parent sets $Pa(m)$; child sample budgets $1 \leq d_m \leq T$.

Ensure: Reconstructed matrix $\hat{X}_i$, or FAILURE.

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1: Initialize  $\hat{X}_i = \mathbf{0}$ .
2: for each DAG  $\mathcal{G}_k$  do
3:   for each metric  $m$  in order  $\pi_k$  do
4:     if  $Pa(m) = \emptyset$  then
5:       Collect all  $T$  values and set  $\hat{X}_i(m, :) = X_i(m, :)$ .
6:     else
7:       Draw  $d_m$  distinct time steps uniformly without replacement; encode them in  $\Omega_m$ .
8:       Set  $D_m = \text{diag}(\Omega_m)$ ; collect  $\mathbf{y}_m = \mathbf{x}_{m,i} D_m$ .
9:       Set  $\hat{P}_m = \hat{X}_i(Pa(m), :)$  and  $A_m = \hat{P}_m D_m$ .
10:      if  $\text{rank}(A_m) < |Pa(m)|$  then
11:        return FAILURE.
12:      end if
13:       $\hat{\mathbf{w}}_m = \mathbf{y}_m A_m^\dagger$ .
14:       $\hat{X}_i(m, :) = \hat{\mathbf{w}}_m \hat{P}_m$ .
15:    end if
16:  end for
17: end for
18: return  $\hat{X}_i$ .
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COROLLARY A.6 (CORRECTNESS OF ALGORITHM 1). *Under the stated exact linear relations, linear independence of parent vectors, and incoherence assumptions, let $N = M - \sum_k l_k \geq 1$ be the number of children and let $0 < \delta < 1$. If Algorithm 1 uses a budget $1 \leq d_m \leq T$ satisfying*

$$d_m \geq \min\{T, \lceil 8\mu\tau_m \log(N\tau_m/\delta) \rceil\} \quad (14)$$

for every child, then it returns $\hat{X}_i = X_i$ with probability at least $1 - \delta$.

PROOF. For analysis, assign every child its uniformly drawn sampling vector in advance, including children not reached if the algorithm stops. This is equivalent to drawing the vectors as they are needed in line 7, since the budgets are fixed and the draws do not depend on sampled values. For each child, define the event $\mathcal{E}_m = \{\text{rank}(\mathbf{P}_m \mathbf{D}_m) = \tau_m\}$ using its true parent matrix $\mathbf{P}_m$. Proposition A.3, with $q = \tau_m$ and $\varepsilon = \delta/N$, gives $\Pr(\mathcal{E}_m^c) \leq \delta/N$; if $d_m = T$, the rank is preserved deterministically. Thus the event $\mathcal{E} = \bigcap_{m \text{ child}} \mathcal{E}_m$ satisfies

$$\Pr(\mathcal{E}) \geq 1 - \sum_{m \text{ child}} \Pr(\mathcal{E}_m^c) \geq 1 - \delta. \quad (15)$$

The union bound requires no independence between these events.

On $\mathcal{E}$, we prove correctness by induction along each order π_k . Every root is collected in full and is therefore exact (line 5). Suppose all metrics preceding a child m have been reconstructed correctly. Since π_k is a causal order, all its parents precede it, so $\widehat{\mathbf{P}}_m = \mathbf{P}_m$ in line 9. The event $\mathcal{E}_m$ makes $\mathbf{A}_m = \mathbf{P}_m \mathbf{D}_m$ full row rank, so the check in line 10 passes. Case 2 of Lemma A.4 then gives $\widehat{\mathbf{w}}_m = \mathbf{w}_m$ and $\widehat{X}_i(m, :) = X_i(m, :)$ in lines 13–14. Induction over all DAGs shows that the algorithm reaches line 18 and returns $\widehat{X}_i = X_i$ on $\mathcal{E}$, which has probability at least $1 - \delta$. $\square$

Each completed run collects $T \sum_k l_k + \sum_{m \text{ child}} d_m$ entries; a run that reports failure may collect fewer. The child budget therefore does not describe the cost of every row. If $N = 0$, all metrics are roots and the algorithm recovers the matrix deterministically by full sampling.

Scope of the comparison. The exact DAG relations imply $r \leq \sum_k l_k \leq Kl$. Thus $Kl \leq r$ is a restrictive sufficient condition, requiring independent root trajectories and equal root counts across DAGs; it does not follow from sparsity alone. Without it, the local and global budgets can still be compared at the same failure tolerance ε using $\tau_m \leq r: 8\mu\tau_m \log(\tau_m/\varepsilon) \leq 8\mu r \log(r/\varepsilon)$. These are sufficient budgets for known subspaces, not necessary sample counts or claims of universal optimality.

Extension to sampled roots. If roots instead use temporal SMC, their recovery events must also be included. Let $q_m = \text{rank}(\mathbf{H}_m)$ for an exactly known historical matrix, and assume the new root lies in its column span. With incoherence at most μ , assign roots the budget $\min\{T, \lceil 8\mu q_m \log(Mq_m/\delta) \rceil\}$ for $q_m \geq 1$, and children the corresponding budget with q_m replaced by τ_m . Proposition A.3, applied to $\mathbf{H}_m^\top$, preserves its sampled rank. Every sampled least-squares solution then gives the same root trajectory, since $\ker(\mathbf{D}_m \mathbf{H}_m) = \ker(\mathbf{H}_m)$. A union bound over at most M recovery events followed by topological induction gives whole-batch recovery with probability at least $1 - \delta$. A zero-rank historical space contains only the zero vector and needs no samples under this membership assumption. Counting only changes of the sampling vector cannot replace this argument: different parent spaces can fail on the same sample set.

Sampling conditions. The results above assume uniform random sampling, exact parent data or correctly reconstructed parents, and the stated linear relations. They do not automatically extend to the equal-interval schedule or to anomalies outside the parent span. Additional samples preserve an already full-rank representation, but do not establish subspace membership by themselves.

B Anomaly-Model Learning and Updating

This appendix derives the approximate EM updates used by the Anomaly Sampler and Model Updater (Sections 4.3 and 4.6). The SCS fixes the allowed source metrics $m' \in \text{Pa}(m) \cup \{m\}$; only their propagation strengths and decay rates are learned. We use the exponential kernel $g_{mm'}(v) = a_{mm'} \beta_{mm'} e^{-\beta_{mm'} v}$ for $v > 0$ and the conditional event rate in Equation (9).

B.1 Likelihood and Event-Source Probabilities

Let h denote the final observed time step: $h = WT$ during offline learning and $h = iT$ after online batch i . The modeled history begins at time 0, and $E_m(h+1)$ includes all detected events of metric m through time h . For the continuous-time Hawkes model evaluated at these event times, the log-likelihood is

$$\ell(\theta) = \sum_{m=1}^M \left[\sum_{s \in E_m(h+1)} \log \lambda_m(s) - \int_0^h \lambda_m(t) dt \right]. \quad (16)$$

The event terms reward rates that explain the detected anomalies, while the integrated rate also accounts for time without events. For the exponential kernel, the integrated rate is

$$\int_0^h \lambda_m(t) dt = u_m h + \sum_{m' \in Pa(m) \cup \{m\}} a_{mm'} B_{mm'}(\beta_{mm'}; h), \quad (17)$$

where the finite-horizon contribution of source metric m' is

$$B_{mm'}(\beta; h) = \sum_{v \in E_{m'}(h+1)} (1 - e^{-\beta(h-v)}). \quad (18)$$

Following the branching interpretation of Hawkes processes [4], each target event has an unobserved source: spontaneous occurrence or triggering by an earlier self or parent event. With the parameters fixed at their current values, the E-step assigns the source probabilities

$$p_{ss}^m = \frac{u_m}{\lambda_m(s)}, \quad p_{sv}^{mm'} = \frac{g_{mm'}(s-v)}{\lambda_m(s)}, \quad v \in E_{m'}(s). \quad (19)$$

The allowed sources satisfy $m' \in Pa(m) \cup \{m\}$, and $E_{m'}(s)$ contains events strictly before s . These probabilities sum to one for each target event when $\lambda_m(s) > 0$.

B.2 Offline Parameter Updates

At each offline iteration, we recompute the source probabilities and the following statistics over the training events:

$$S_{u_m} = \sum_{s \in E_m(h+1)} p_{ss}^m, \quad S_{a_{mm'}} = \sum_{s \in E_m(h+1)} \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}. \quad (20)$$

$$S_{\beta_{mm'}} = \sum_{s \in E_m(h+1)} \sum_{v \in E_{m'}(s)} p_{sv}^{mm'} (s-v). \quad (21)$$

Here S_{u_m} is the expected number of spontaneous events, $S_{a_{mm'}}$ is the expected number triggered by metric m' , and $S_{\beta_{mm'}}$ is their probability-weighted total triggering delay.

Holding these probabilities fixed, the expected complete-data log-likelihood separates into a background term $Q_{u_m} = S_{u_m} \log u_m - u_m h$ and, for each allowed source-target pair,

$$Q_{mm'} = S_{a_{mm'}} (\log a_{mm'} + \log \beta_{mm'}) - \beta_{mm'} S_{\beta_{mm'}} - a_{mm'} B_{mm'}(\beta_{mm'}; h). \quad (22)$$

Terms independent of the updated parameters are omitted. The background update is $u_m = S_{u_m}/h$. For an identifiable source-target pair, the exact finite-horizon stationary conditions are

$$a_{mm'} = \frac{S_{a_{mm'}}}{B_{mm'}(\beta_{mm'}; h)}, \quad (23)$$

$$\frac{S_{a_{mm'}}}{\beta_{mm'}} - S_{\beta_{mm'}} - a_{mm'} \sum_{v \in E_{m'}(h+1)} (h-v) e^{-\beta_{mm'}(h-v)} = 0. \quad (24)$$

Thus, exact finite-horizon updates couple the propagation strength and decay rate.

CauSample uses a boundary approximation to obtain explicit updates. Writing $n_{m'} = |E_{m'}(h+1)|$, we replace $B_{mm'}(\beta; h)$ by $n_{m'}$ and omit its derivative with respect to β . With the E-step statistics fixed, the approximate objective is $\tilde{Q}_{mm'} = S_{a_{mm'}}(\log a_{mm'} + \log \beta_{mm'}) - \beta_{mm'} S_{\beta_{mm'}} - a_{mm'} n_{m'}$. Setting its first derivatives, and that of Q_{u_m} , to zero gives

$$\frac{S_{u_m}}{u_m} - h = 0, \quad \frac{S_{a_{mm'}}}{a_{mm'}} - n_{m'} = 0, \quad \frac{S_{a_{mm'}}}{\beta_{mm'}} - S_{\beta_{mm'}} = 0. \quad (25)$$

For positive, identifiable statistics these objectives have negative second derivatives, so the solutions maximize the approximate objective. The resulting approximate M-step is

$$u_m = \frac{S_{u_m}}{h}, \quad a_{mm'} = \frac{S_{a_{mm'}}}{n_{m'}}, \quad \beta_{mm'} = \frac{S_{a_{mm'}}}{S_{\beta_{mm'}}}. \quad (26)$$

These ratios apply when the denominators are positive and the event statistics identify the corresponding parameters. For a source–target pair with no positive triggering responsibility, $S_{a_{mm'}} = S_{\beta_{mm'}} = 0$; its decay rate is then unidentifiable, and the corresponding ratio does not apply. The approximation requires both the omitted kernel tails $\sum_v e^{-\beta_{mm'}(h-v)}$ and their derivative contributions to be small relative to the retained terms. A long observation horizon alone does not ensure this condition when many source events occur near its end. Consequently, these are approximate EM updates, rather than exact M-steps for the finite-horizon objective in Equation (22). We repeat the offline E- and M-steps with $h = WT$ until the maximum parameter change falls below ϵ .

B.3 Incremental Updates after Each Batch

We initialize the cumulative statistics from the final offline E-step. After batch i , each newly sampled anomaly at time s in metric m undergoes one E-step followed by one approximate M-step. Its source probabilities are computed using the current parameters and only events strictly before s , then added to the cumulative statistics:

$$S_{u_m} \leftarrow S_{u_m} + p_{ss}^m, \quad S_{a_{mm'}} \leftarrow S_{a_{mm'}} + \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}, \quad (27)$$

$$S_{\beta_{mm'}} \leftarrow S_{\beta_{mm'}} + \sum_{v \in E_{m'}(s)} p_{sv}^{mm'}(s-v). \quad (28)$$

For $m' \in Pa(m) \cup \{m\}$, we update the corresponding parameters using Equation (26) with $h = iT$ and $n_{m'} = |E_{m'}(iT+1)|$. Both quantities include the initial training period and all subsequent batches. They remain fixed while processing the new events from the current batch; each parameter update precedes the E-step for the next event. The resulting parameters are used for predictions in the next batch.

Past source probabilities remain fixed, including contributions from batches that have left the sliding window. This cumulative update avoids repeated EM iterations over old target events, although each new target event still uses earlier parent and self events to compute its source probabilities. It does not claim exact maximization or monotonic improvement of the full-history likelihood. If a batch contains no sampled anomalies, no parameter update is performed; its elapsed time is included when the next event triggers an update. Within-batch predictions incorporate new events with fixed parameters, whereas parameter learning takes place after the batch ends.

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